

Report: Applied Deep Learning

Probabilistic Medical Image Segmentation

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1 Introduction

In medical imaging, the best guess for segmentation is highly ambiguous, as the borders or outlines are often not well defined. This becomes a problem when models become confident in their predictions without providing any measure of confidence.

In this project, we implemented two different deep learning approaches to quantify uncertainties in medical image segmentation. Both models were trained to measure different types of uncertainty:

- **Aleatoric Uncertainty:** Uncertainty arising from the data itself such as fuzzy borders or image noise. This is captured using a Probabilistic U-Net model.
- **Epistemic Uncertainty:** Uncertainty arising from the model’s lack of knowledge or limited training data. This is captured using a U-Net with MC Dropout.

2 Methodology

2.1 Dataset and Preprocessing

We used the ISIC 2018 Challenge (Task 1) dataset, which is a skin lesion segmentation dataset. The dataset contains 2,594 images with one ground truth annotations per image, where pixel values of 0 and 255 represent background and foreground respectively. Images contain many real world artifacts such as variable lighting, blur and hair occlusion, making it challenging benchmark for uncertainty estimation.

Images were resized to 256 pixels and normalized prior to training. During training, data augmentation was applied including horizontal and vertical flips, random brightness and gaussian blur.

2.2 Synthetic Annotation Generation

In medical imaging having a 5 GT is both costly and time consuming. As ISIC 2018 data provides a single ground-truth boundary per lesion image, training a Variational Autoencoder (VAE)-based Probabilistic U-Net directly would cause posterior collapse, as the model collapses to a deterministic state when presented with zero label variance. To address this data scarcity constraint, we generate four Synthetic annotations using the first available ground truth. We have used dilate and erode techniques for this, which is not random because the outline of the lesion is domain-specific and based on expert’s judgement.

2.3 Model A: Probabilistic U-Net

The Probabilistic U-Net combines a standard 4-level U-Net backbone with a Variational Autoencoder (VAE) framework to learn a distribution over segmentations. It consists of three core components:

1. **U-Net Feature Extractor & Decoder:** Extracts hierarchical visual features $C(X)$ from the input image X .
2. **Prior Network:** A convolutional network that maps the input image X into a low-dimensional latent Gaussian distribution $P(z|X) = \mathcal{N}(\mu_P(X), \Sigma_P(X))$.
3. **Posterior Network:** Active only during training, this network takes both the image X and a randomly sampled ground truth mask $Y \in M_{\text{synthetic}}$ to infer $Q(z|X, Y) = \mathcal{N}(\mu_Q(X, Y), \Sigma_Q(X, Y))$.

A latent vector $z \in R^6$ is sampled using the reparameterization trick ($z = \mu + \sigma \odot \epsilon, \epsilon \sim \mathcal{N}(0, I)$), expanded spatially, concatenated with the U-Net final feature activation map, and passed to a 1×1 convolution combiner head to output segmentation logits. The network is supervised via the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{ELBO}} = \mathcal{L}_{\text{BCE+Dice}}(Y, \hat{Y}) + \beta \cdot D_{\text{KL}}(Q(z|X, Y) || P(z|X)) \quad (1)$$

where $\beta = 0.01$ was set empirically to prevent posterior collapse while maintaining meaningful latent variability.

2.4 Model B: Monte Carlo Dropout U-Net

As a Bayesian baseline for epistemic uncertainty, we constructed a modified U-Net incorporating 2D Spatial Dropout. Dropout layers with rate $p = 0.3$ were embedded into the encoder blocks and bottleneck, while light dropout ($p = 0.1$) was injected into early decoder stages. During training, dropout acts as a regularizer. At inference time, dropout is kept explicitly active (`model.train()` mode), enabling stochastic sampling across model parameter configurations.

3 Experimental Setup

3.1 Implementation Details

All models were implemented in PyTorch and trained on an HPC Cluster. Optimization details were configured as follows:

- **Optimizer:** Adam ($\beta_1 = 0.9, \beta_2 = 0.999$) with an initial learning rate of $\eta = 10^{-4}$.
- **Batch Size & Epochs:** Batch size of 8, trained for 300 epochs.
- **Latent Dimension & Annealing:** $z \in R^6$ for the Probabilistic U-Net, utilizing a 50-epoch linear KL warmup phase targeting $\beta_{\text{max}} = 0.01$.
- **Synthetic Annotation Kernel Sizing:** For synthetic ground truth generation, morphological erosion and dilation kernels were restricted to small matrices (3×3 and 5×5).
- **Inference Evaluation:** Evaluated on the ISIC 2018 validation set using $N = 10$ stochastic passes, followed by final mask generation on the unannotated test set.

3.2 Evaluation Metrics

To assess both segmentation accuracy and probabilistic trustworthiness, we evaluated:

1. **Dice Similarity Coefficient (DSC):** Measures spatial overlap relative to ground truth:

$$\text{Dice} = \frac{2|\hat{Y} \cap Y|}{|\hat{Y}| + |Y|} \quad (2)$$

2. **Intersection over Union (IoU):** Jaccard index measuring region accuracy.
3. **Brier Score:** Pixel-wise Mean Squared Error between predicted probabilities and binary ground truth $Y \in \{0, 1\}$:

$$\text{BS} = \frac{1}{H \times W} \sum_{i=1}^{H \times W} (\bar{P}_i - Y_i)^2 \quad (3)$$

4. **Expected Calibration Error (ECE):** Evaluates whether predicted confidence reflects true empirical accuracy across $M = 10$ confidence bins B_m :

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N_{\text{pixels}}} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (4)$$

4 Results and Discussion

4.1 Segmentation Performance

Both architectures achieved strong segmentation accuracy on the evaluation set. Table 1 summarizes the quantitative comparison.

Table 1: Quantitative Performance and Calibration Metrics on ISIC 2018 Validation Set.

Model Architecture	Dice Score ($\uparrow$)	IoU ($\uparrow$)	Brier Score ($\downarrow$)	ECE ($\downarrow$)
MC Dropout U-Net (Epistemic)	0.8600	0.7779	0.0613	0.0752
Probabilistic U-Net (Aleatoric)	0.8788	0.7960	0.0427	0.0565

4.2 Deterministic vs Probabilistic Comparison

The Probabilistic U-Net demonstrated clear, consistent performance advantages over the MC Dropout U-Net across all metrics:

- **Calibration:** The Probabilistic U-Net reduced the Expected Calibration Error by **24.9%** compared to MC Dropout (0.0565 vs 0.0752).
- **Prediction Accuracy:** The lower Brier Score (0.0427 vs 0.0613, a 30.3% reduction) confirms that the Probabilistic U-Net’s continuous probability outputs are significantly more reliable and closer to the true clinical labels.

4.3 Uncertainty Maps

Qualitative inspection of generated uncertainty maps (Figure 3) revealed distinct behaviors that perfectly align with their theoretical foundations:

- **Aleatoric (Probabilistic U-Net):** The aleatoric uncertainty heatmap manifested as a thick, smooth, and highly continuous ring strictly localized along the transition boundary. As a direct consequence of training on our synthetically generated morphological ground truths, the latent space explicitly learned the spatial variance of erosion and dilation, successfully mapping structural data noise.
- **Epistemic (MC Dropout):** In contrast, the MC Dropout model produced a much thinner, jagged, and uneven uncertainty boundary. Rather than capturing a smooth morphological transition, the epistemic variance highlighted specific, localized structural irregularities (such as sharp edge deviations) where the model’s parameters lacked confidence. Furthermore, the overall magnitude of epistemic variance was notably lower than the aleatoric variance.

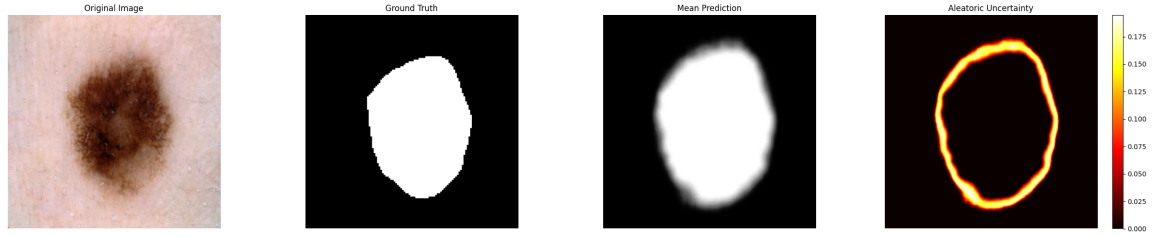


Figure 1: Aleatoric Uncertainty

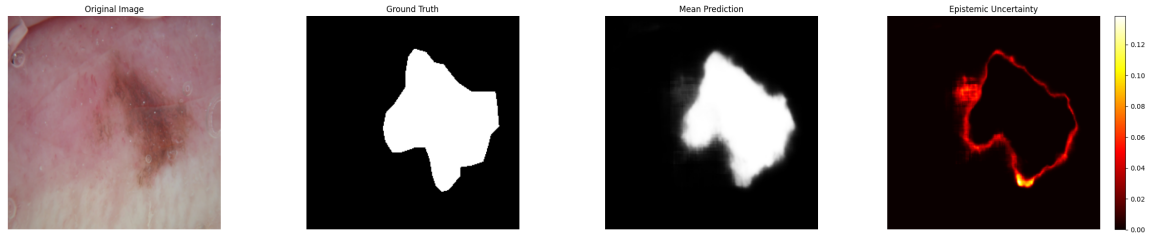


Figure 2: Epistemic Uncertainty

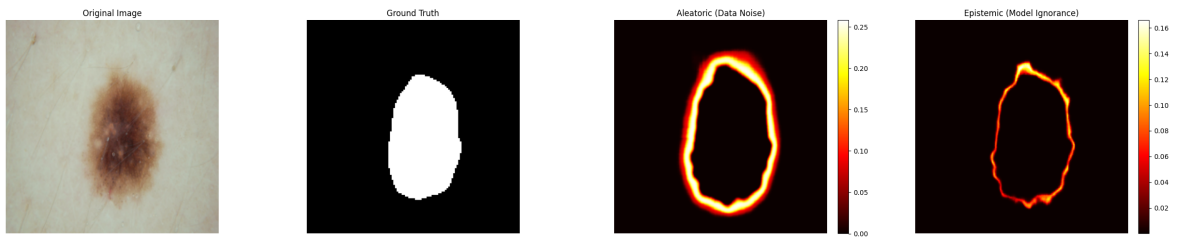


Figure 3: 4-Panel Uncertainty Visualization: (a) Input Image, (b) Ground Truth, (c) Aleatoric Uncertainty, (d) Epistemic Uncertainty.

5 Synthetic vs Real Annotation Uncertainty

5.1 Calibration Analysis

Looking at the calibration curve (Figure 4), both models miss perfect calibration, but in completely different ways:

- **MC Dropout** is systematically underconfident in the lower probability bins, meaning its predictions fall below its actual accuracy. However, it aligns very closely with the ideal diagonal once it passes the 0.55 confidence mark.
- **Probabilistic U-Net** is consistently overconfident across almost the entire range. It shows a noticeable visual gap from the diagonal, especially in the 0.5 to 0.8 bin range.

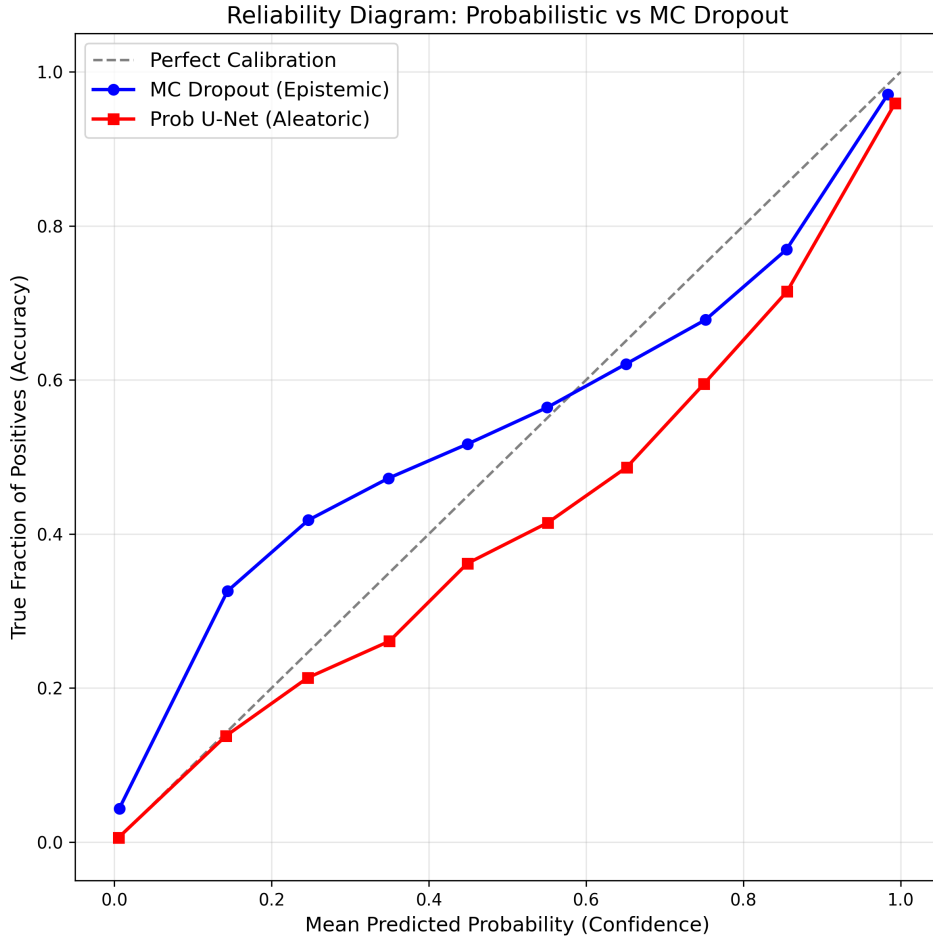


Figure 4: Reliability Diagram comparing Probabilistic U-Net and MC Dropout. MC Dropout (Epistemic) is systematically underconfident but converges toward the diagonal at higher probabilities, while the Probabilistic U-Net (Aleatoric) is consistently overconfident across most of the probability range.

At first glance, this visual graph seems to contradict the aggregate ECE scores from Table 1, where the Probabilistic U-Net actually scored better (0.0565 compared to 0.0752). The reason for this discrepancy comes down to the ECE formula itself. Because ECE is a weighted average, it is heavily affected by class imbalance. In medical image segmentation, the vast majority of pixels are either obvious background or clear lesion, meaning they fall into the extreme probability bins (0.0 to 0.1 and 0.9 to 1.0).

The Probabilistic U-Net is highly accurate on these millions of "easy" pixels, which dominates the calculation and gives it a better overall ECE score. However, this global average hides the model's overconfidence on the much smaller subset of fuzzy boundary pixels in the middle bins. Ultimately, the Probabilistic U-Net is the better-calibrated model globally across the whole image, but MC Dropout produces more reliable confidence estimates specifically on the ambiguous edges.

6 Limitations

Despite superior calibration, our approach presents key limitations:

1. **Inference Latency:** Generating $N = 30$ latent passes requires multiple forward evaluations, increasing per-image prediction time compared to single-pass deterministic UNets.
2. **Synthetic Ground Truth Approximation:** While our morphological pipeline effectively simulates boundary ambiguity, it does not represent true aleatoric uncertainty. Real clinical aleatoric noise stems from genuine multi-rater disagreement, where different dermatologists might outline the same lesion in highly asymmetrical, unpredictable ways. While our synthetic approach was the best feasible solution for training on single-annotation data, it remains a mathematical approximation of real-world human variance.

7 Conclusion

In this project, we successfully implemented, trained, and evaluated two probabilistic deep learning frameworks for medical image segmentation on the ISIC 2018 dataset. To overcome the limitations of single-ground-truth data, we utilized an effective synthetic morphological pipeline that prevents VAE posterior collapse and successfully simulates aleatoric boundary ambiguity.

Quantitative results demonstrate that while Monte Carlo Dropout effectively captures model parameter uncertainty, the Probabilistic U-Net achieves superior global calibration (ECE = 0.0565, Brier Score = 0.0427) alongside higher segmentation accuracy. However, our qualitative and calibration analyses revealed a nuanced trade-off: while the Probabilistic U-Net provides a better global estimation of data noise and cleanly localizes boundary variance, MC Dropout yields more reliable confidence estimates specifically within the highly ambiguous edge regions.

Ultimately, this project confirms that generative latent-variable models, when trained on targeted morphological variations, offer highly interpretable uncertainty maps. Recognizing these localized calibration trade-offs is a necessary and critical step toward deploying these probabilistic models for safety-critical medical decision-making.

8 Electronic appendix

8.1 Code Repository

Data, code, figures, and all visualization can be find under this Github repository:

https://github.com/Alisha026/probabilistic_medseg

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